

Eyal Itskovits

Curriculum Vitae

Profile

Experienced computational scientist and manager.

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Professional Experience

2024-Current

Director of Machine Learning and Artificial Intelligence

Leading GSK Flagship AI Scientist project

Leading a multi-team, globally distributed project, with direct reporting and engagement with GSK's Global Leadership Team and Board.

2021-2024

Senior AIML Research Engineer and Team Leader
Team leading, AIML Reserch

Lead a team of five PhD-level researchers applying cutting-edge machine learning to solve challenging problems in oncology

2007-2009

Developer at OpTier

Software development and reverse engineering
Information security related product development and reverse engineering.

Academic experience

March 2013 – January 2020

PhD Candidate

Prof. Alon Zaslaver's Group: I have been working on research questions dealing with neural encoding of complex stimuli, and with learning and memory in compact neural networks (See publications section).

Education

2015-2020 Computer Science and Computational Biology, PhD
Systems Neuroscience

2013-2015 Computer Science and Computational Biology, MSc
gpa: 93

2009-2012 Computer Science and Computational Biology, BSc
gpa: 90

Skills

Data Analysis, Machine Learning and Software Engineering

Practical and theoretical experience with:

- Statistical machine learning.
- Deep-learning.
- Signal processing, and time-series analysis.
- Strong mathematical and statistical background.

Software Development

- Experience with design and implementation of small and large software projects
- Proficient in C, C++, Python. Experience with server, desktop and web applications.
- Experience with operation systems internals. A strong expertise with Linux environments and cloud computing.

Life Sciences

- Experience with various molecular biology procedures.
- Experience with advanced optical systems.
- Analysis of high-dimensional neural activity data, and next-generation RNA Sequencing.

Teaching Experience

- 2014-2019 Probability and Statistics for the Life Sciences
Hebrew University - Course 72301
- 2013 OOP - Object Oriented Programming
Hebrew University - Course 67125
- 2012 General Genetics
Hebrew University - Course 72155

Professional Service

Journal Editorial Service

- Reviewing Editor, eLife – Board of Reviewing Editors, 2026–present

Conference Reviewer

- Neural Information Processing Systems (NeurIPS), 2024
- International Conference on Learning Representations (ICLR) – ML For Drug Discovery Workshop, 2023

Honors and Awards

- UK Global Talent Endorsement (Exceptional Promise) - Royal Academy of Engineering - 2026
- London Tech Week 2026 - Invited Speaker
- JBC Postdoctoral Fellowship - 2020
- HHMI Janelia's Junior Scientist Workshop on Machine Learning and Computer Vision participant - 2019
- Next-Gen Neuro Workshop for Advanced PhD Students participant - 2019
- The Irene and Michael Mochrik Award For Outstanding Genetics Students - 2018
- SMART Award, Alexander Silverman Institute of Life Sciences - 2018
- Excellent Teaching Assistant - 2013-2016, 2018-2019
- JBC Golden PhD Scholarship - 2017
- Best Poster - I-Core Single Molecule Biophysics Conference - 2015
- JBC Travel Scholarship - 2015

- Honorable Mention for poster - International Worm Meeting - 2015
- Mada - Life Sciences Excellence Program - 2013
- Dean's List - 2010-2012, 2013-2015

Previous academic experience

October 2013 – March 2013

MSc Student

Rotational Student

Prof. Sagiv Shifman's Lab - I analysed large-scale datasets of individuals with autism spectrum disorder (ASD) in search of the genomic correlates of the syndrome.

Prof. Nir Friedman's Lab - I studied a novel gene-regulation mechanism.

October 2011 - October 2012

BSc Student

Movement Ecology Lab

I used machine learning to classify animal behavior using accelerometer temporal signal.

Volunteering

2016-2017 "Aleh" Running Team - Biweekly escorting of blind and visually impaired runners

2015-2017 "Magdilim Rosh" - Exposing high school students to neuroscience research

2014 "Exposure to Science" - Introducing research and basic science topics to adolescents from complex backgrounds

2011 "Magshimim Program" - Teaching computer science and information security to high school students from the social periphery

Online profiles

- [Google Scholar](#)
- [LinkedIn](#)
- [Twitter](#)
- [GitHub](#)
- [StackOverflow](#)

Publications

- Deshe, Noa, Yifat Eliezer, Lihi Hoch, Eyal Itskovits, Shachaf Ben-Ezra, and Alon Zaslaver. n.d.
"Inheritance of Associative Memories in *C. Elegans* Nematodes."
BioRxiv, <https://doi.org/10.1101/2020.09.08.287318>.
- Pritz, Christian O., Eyal Itskovits, Eduard Bokman, Rotem Ruach, Vladimir Gritsenko, Tal Nelken, Mai Menasheerof, Aharon Azulay, and Alon Zaslaver. n.d.
"Principles for Coding Associative Memories in a Compact Neural Network."
BioRxiv, <https://doi.org/10.1101/2020.06.20.162818>.
- Ishtayeh, Hasan, Hanna Achache, Eitan Kroizer, Yisrael Rappaport, Eyal Itskovits, Hila Gingold, Corinne Best, Oded Rechavi, and Yonatan B. Tzur. 2020.
"Systematic Analysis of Long Intergenic Non-Coding RNAs in Germline Uncovers Roles in Somatic Growth."
RNA Biology, September, 1–11.
- Iwanir, Shachar, Rotem Ruach, Eyal Itskovits, Christian O. Pritz, Eduard Bokman, and Alon Zaslaver. 2019.
"Irrational Behavior in *C. Elegans* Arises from Asymmetric Modulatory Effects within Single Sensory Neurons."
Nature Communications 10 (1): 3202.
- Itskovits, Eyal, Rotem Ruach, and Alon Zaslaver. 2018.
"Concerted Pulsatile and Graded Neural Dynamics Enables Efficient Chemotaxis in *C. Elegans*."
Nature Communications 9 (1): 2866.
- Mukherjee, Diptendu, Bogna Marta Ignatowska-Jankowska, Eyal Itskovits, Ben Jerry Gonzales, Hagit Turm, Liz Izakson, Doron Haritan, et al. 2018.
"Salient Experiences Are Represented by Unique Transcriptional Signatures in the Mouse Brain."
eLife 7 (February). <https://doi.org/10.7554/eLife.31220>. (Joint first author)
- Itskovits, Eyal, Amir Levine, Ehud Cohen, and Alon Zaslaver. 2017.
"A Multi-Animal Tracker for Studying Complex Behaviors."
BMC Biology 15 (1): 29.
- Azulay, Aharon, Eyal Itskovits, and Alon Zaslaver. 2016.
"The *C. Elegans* Connectome Consists of Homogenous Circuits with Defined Functional Roles."
PLoS Computational Biology 12 (9): e1005021.
- Bean, Angela, Assaf Weiner, Amanda Hughes, Eyal Itskovits, Nir Friedman, and Oliver J. Rando. n.d.
"Genome-Wide Histone Modification Patterns in *Kluyveromyces Lactis* Reveal Evolutionary Adaptation of a Heterochromatin-Associated Mark."
BioRxiv, <https://doi.org/10.1101/039776>.